Práctica 5. Primera aproximación a Docker

Añadir la clave GPG oficial de Docker

```
david@Despliegue:~$ sudo apt install ca-certificates curl
Upgrading:
  ca-certificates  curl  libcurl3t64-gnutls  libcurl4t64

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 63
  Download size: 1,261 kB
  Space needed: 4,096 B / 21.0 GB available

Continue? [Y/n] y^[S
```

```
david@Despliegue:~$ sudo install -m 0755 -d /etc/apt/keyrings
david@Despliegue:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
david@Despliegue:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
david@Despliegue:~$ _
```

Añadir el repositorio a las fuentes de APT

```
david@Despliegue:~$ sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF
[sudo: authenticate] Password:
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: resolute
Components: stable
Architectures: amd64
Signed-By: /etc/apt/keyrings/docker.asc
```

Instalar los paquetes de Docker

```
david@Despliegue:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
[sudo: authenticate] Password:
Installing:
  containerd.io  docker-buildx-plugin  docker-ce  docker-ce-cli  docker-compose-plugin

Installing dependencies:
  docker-ce-rootless-extras  pigz

Suggested packages:
  cgroupfs-mount  | cgroup-lite  docker-model-plugin

Summary:
  Upgrading: 0, Installing: 7, Removing: 0, Not Upgrading: 2
  Download size: 102 MB
  Space needed: 388 MB / 21.0 GB available

Continue? [Y/n] y
Get:1 http://es.archive.ubuntu.com/ubuntu resolute/universe amd64 pigz amd64 2.8-1build1 [66.8 kB]
Get:2 https://download.docker.com/linux/ubuntu resolute/stable amd64 containerd.io amd64 2.4.2-3~ubuntu.26.04~resolute [
23.3 MB]
Get:3 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-cli amd64 5:29.9.2-1~ubuntu.26.04~resolut
e [17.6 MB]
Get:4 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce amd64 5:29.9.2-1~ubuntu.26.04~resolute [2
4.6 MB]
Get:5 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-buildx-plugin amd64 0.38.1-1~ubuntu.26.04~re
solute [17.8 MB]
Get:6 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-rootless-extras amd64 5:29.9.2-1~ubuntu.2
6.04~resolute [10.2 MB]
Get:7 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-compose-plugin amd64 5.6.1-1~ubuntu.26.04~re
solute [8,085 kB]
```

Después de la instalación, verifica que Docker esté en funcionamiento

```
david@Despliegue:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-22 16:10:54 UTC; 1min 23s ago
 Invocation: 53e3e98f7bb34e8889525c4e2b933f5e
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 18126 (dockerd)
      Tasks: 10
     Memory: 26.8M (peak: 27.2M)
        CPU: 343ms
     CGroup: /system.slice/docker.service
             └─18126 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.519933554Z" level=info msg="Restoring containers: >
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.535157471Z" level=info msg="Deleting nftables IPv4>
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.550412105Z" level=info msg="Deleting nftables IPv6>
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.720734602Z" level=info msg="Loading containers: do>
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.724783419Z" level=info msg="Docker daemon" commit=>
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.724884826Z" level=info msg="Initializing buildkit"
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.865445863Z" level=info msg="Completed buildkit ini>
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.870207189Z" level=info msg="Daemon has completed i>
Oct 22 16:10:54 Despliegue dockerd[18126]: time="2026-10-22T16:10:54.870403801Z" level=info msg="API listen on /run/doc>
Oct 22 16:10:54 Despliegue systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-23/23 (END)
```

Ejecutar docker run --name mi-apache -p 8080:80 -d httpd y comprobar

```
david@Despliegue:~$ sudo docker run --name mi-apache -p 8080:80 -d httpd
[sudo: authenticate] Password:
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
b361b1ecdecd: Pull complete
f2be67225471: Pull complete
1854ebd7c33a: Pull complete
4f4fb700ef54: Pull complete
f8903be41a4f: Pull complete
720ea190dd65: Pull complete
8d61ccc1b972: Download complete
096bb8816822: Download complete
Digest: sha256:a30cd4cd57f3f3c3be9b2d16384d2488839996bb14c67df2b51f1ecbb3e584a1
Status: Downloaded newer image for httpd:latest
1b3fe63039296a27007987e8a70da2079d264384c32376c7300511c254bd1631
david@Despliegue:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-22 16:46:19 UTC; 1min 33s ago
 Invocation: 9a7e404463c049fb8040ad69c5724e20
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 1210 (dockerd)
      Tasks: 25
     Memory: 167.4M (peak: 167.4M)
        CPU: 918ms
     CGroup: /system.slice/docker.service
             ├─1210 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
             └─1744 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 8080 -container-ip 172.17
```

Comprobar

